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Bacillus toyonensis biovar Thuringiensis Bto_UNVM-42: A novel strain with potential for the biological control of nematodes

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ABSTRACT

Bacillus toyonensis is a gram-positive bacterium within the *Bacillus cereus* group, primarily recognized for its probiotic properties. Recently, a strain designated *B. toyonensis* biovar Thuringiensis (Bto_UNVM-94) was reported to produce parasporal crystals with dual insecticidal activity against *Cydia pomonella* and *Anthonomus grandis*. In this study, we describe and characterize a previously undescribed strain, *B. toyonensis* biovar Thuringiensis Bto_UNVM-42, which exhibits nematicidal and nematostatic activity, representing the first report of these effects within this species. Whole-genome sequencing revealed the presence of a large plasmid containing multiple putative pesticidal genes, including five *cry* genes encoding Cry32-like proteins, two additional *cry* genes encoding Cry69-like and Cry73-like proteins, three *cyt* genes encoding Cyt1-like proteins, and two *mpp* genes encoding Mpp-like proteins. In addition, the genome encodes four chitinases and two proteins with significant similarity to the nematicidal Sep1 protease from *Bacillus firmus*. Proteomic analysis of parasporal crystals confirmed the expression of Cry32-family proteins. Bioassays showed that strain Bto_UNVM-42 was not toxic to *Aedes aegypti*, *Alphitobius diaperinus*, *Anthonomus grandis*, *Cydia pomonella* or *Drosophila suzukii*. In contrast, it exhibited nematicidal activity against *Panagrellus redivivus*, low but detectable mortality (8%) against *Meloidogyne javanica*, and a reversible nematostatic effect on *Pratylenchus zeae*. Taken together, the available bioassay observations and the genomic and proteomic data are consistent with a predominantly enzyme-driven rather than a classical crystal toxin–mediated mode of action. Overall, these findings expand the known biocidal spectrum of *B. toyonensis* and highlight its potential relevance for the development of biological strategies for nematode management.

1. Introduction

Plant-parasitic nematodes, such as root-lesion (*Pratylenchus* spp.) and root-knot (*Meloidogyne* spp.), cause severe economic losses by damaging root systems and reducing crop yields worldwide (Jones et al., 2013; Nicol et al., 2011). Among them, *Meloidogyne javanica* (Tylenchida: Meloidogynidae) is particularly harmful due to its ability to induce gall formation in roots, thereby impairing water and nutrient uptake (Moens et al., 2009). Although chemical nematicides have traditionally been the primary control strategy, their environmental impact, regulatory restrictions, and the emergence of resistant nematode populations highlight the urgent need for sustainable and environmentally friendly alternatives (Danchin et al., 2013). In this context, biological control agents based on *Bacillus* spp. (e.g., *Bacillus thuringiensis*)

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have attracted increasing interest as potential alternatives to chemical nematicides. However, some *B. thuringiensis* strains are known to produce heat-stable metabolites, such as type I β-exotoxin, which exhibit broad-spectrum toxicity and are therefore considered undesirable in biological control agents due to their lack of target specificity and associated environmental and regulatory concerns.

Bacillus toyonensis is a gram-positive, spore-forming bacterium belonging to the *Bacillus cereus* group and has been widely used as a probiotic in animal nutrition, notably as the active component of TOYOCERIN® (Jiménez et al., 2013). Beyond its probiotic use, an insecticidal strain of this species was recently described as *B. toyonensis* biovar Thuringiensis strain Bto_UNVM-94, following a revised taxonomic framework for the *B. cereus* group (Carroll et al., 2020; Sauka et al., 2022). This biovar is defined by its capacity to produce bipyramidal parasporal crystals, a hallmark of *B. thuringiensis*. Spore-crystal mixtures from strain Bto_UNVM-94 showed dual insecticidal activity against *Cydia pomonella* (Lepidoptera: Tortricidae) and *Anthonomus grandis* (Coleoptera: Curculionidae), but no activity was detected against the free-living nematode *Panagrellus redivivus* (Sauka et al., 2022).

For decades, *B. thuringiensis* has been the most widely used microbial agent with specific biocidal activity against a broad range of invertebrates, including nematodes (Palma et al., 2014). Nematicidal activity in *B. thuringiensis* has been mainly attributed to a limited subset of crystal proteins, including Cry5, Cry12, Cry13, Cry14, Cry21, Cry55 and App6 (formerly Cry6), which exert toxicity upon ingestion (Berry et al., 2025; van Frankenhuyzen, 2009; van Frankenhuyzen and Tonon, 2013). However, some plant-parasitic nematodes, particularly sedentary endoparasites such as *Meloidogyne* spp., can evade the effects of these proteins due to their specialized feeding behavior. These nematodes induce the formation of giant cells within plant tissues, which provide a protected nutritional niche and limit direct exposure to orally ingested toxins (Zhang et al., 2012). Accordingly, differences in feeding habits have been shown to influence susceptibility to *B. thuringiensis*-derived proteins, as exemplified by the contrasting toxicity of App6Aa2 (formerly Cry6Aa2) toward the bacterivorous nematode *Caenorhabditis elegans* and the plant-parasitic *Meloidogyne hapla* (Yu et al., 2015).

Alternative biological control strategies based on non-*B. thuringiensis* species have therefore attracted increasing attention. *Bacillus firmus* strain I-1582, for instance, exhibits activity against *Meloidogyne incognita* and has been approved in Europe as a biological nematicide (Ghahremani et al., 2020). In this case, toxicity has been associated with the serine protease Sep1 (peptidase S8 family), which degrades intestinal tissues and exerts nematicidal effects against both *C. elegans* and *M. incognita*, although the precise mode of action remains incompletely understood (Geng et al., 2016).

In this study, we report the genome sequencing and biocidal characterization of a previously undescribed *B. toyonensis* biovar Thuringiensis strain, Bto_UNVM-42, and provide the first evidence of nematicidal and nematostatic activity within this species. Genome analysis revealed the presence of multiple genes encoding insecticidal-like proteins, including several Cry-, Cyt-and MPP-like proteins, as well as additional putative virulence factors such as chitinases and two Sep1 homologs. Proteomic analysis of the excised SDS-PAGE band from parasporal crystals identified Cry32-family proteins. Insect bioassays showed that strain Bto_UNVM-42 was not toxic to *Aedes aegypti* (Diptera: Culicidae), *Alphitobius diaperinus* (Coleoptera: Tenebrionidae), *Anthonomus grandis* (Coleoptera: Curculionidae), *Cydia pomonella* (Lepidoptera: Tortricidae), or *Drosophila suzukii* (Diptera: Drosophilidae). In contrast, the strain exhibited nematicidal activity against the free-living nematode *P. redivivus*, causing complete mortality accompanied by pronounced pathological alterations, showed low but detectable mortality against *M. javanica* (8%) and induced a reversible nematostatic effect on the plant-parasitic nematode *Pratylenchus zeae*.

Taken together, these results identify *B. toyonensis* biovar Thuringiensis strain Bto_UNVM-42 as a bacterium whose nematicidal and nematostatic activities are most consistent with an enzyme-driven rather than toxin-mediated mode of action. This finding expands the functional diversity of the *B. cereus* group and highlights alternative mechanisms that may contribute to the biological control of nematodes.

2. Materials and methods

2.1. Strain isolation and characterization

Soil samples were collected using a soil sampling tube from a native forest area in the province of Jujuy (Argentina). The final sample consisted of a composite of five randomly collected subsamples, totaling approximately 200 g of soil. Samples were transported to the laboratory and stored at 4 °C in zip-lock bags until further processing for bacterial isolation.

Bacterial colonies originating from sporulated bacteria were isolated on CCY agar plates as previously described (Iriarte et al., 1998). The production and morphology of parasporal crystals were examined by light microscopy after Coomassie blue slide staining (Ammons et al., 2002) and further confirmed by scanning electron microscopy (SEM), performed at the Comprehensive Center for Electron Microscopy (CIME-CONICET, Argentina). Caseinolytic activity was assessed using a casein hydrolysis assay as previously described (Geng et al., 2016). The strain was purified to axenic culture, sporulated, and preserved in the laboratory collection at −80 °C in 15% (v/v) glycerol.

2.2. Genome sequencing and annotation

Total genomic DNA was extracted using the Wizard Genomic DNA Purification Kit (Promega, Madison, WI, USA), following the manufacturer's protocol for gram-positive bacteria. DNA integrity was assessed by electrophoresis on 1% agarose gels stained with SYBR Safe (Thermo Fisher Scientific, Waltham, MA, USA), and DNA concentration was quantified using a Picodrop PICO 100 spectrophotometer. Purified DNA was used to construct an Illumina sequencing library, which was sequenced at the Genomics Unit of the National Institute of Agricultural Technology (INTA, Argentina) using high-throughput Illumina technology. Long-read sequencing was performed using Oxford Nanopore technology (Cucare Diagnostics) to obtain long DNA fragments, thereby improving genome assembly contiguity and completeness.

Raw Illumina and Nanopore reads were quality-checked and trimmed prior to hybrid *de novo* assembly using Unicycler (Wick et al., 2017), which integrates short-read accuracy with long-read scaffolding. This approach enabled the closure of the chromosome and associated plasmids. Assembled replicons were initially screened using BLAST (Altschul et al., 1990) against a customized, non-redundant insecticidal protein database (2025 update) curated in our laboratory, and candidate hits were subsequently confirmed using BestMatchFinder in the BPPRC database (Panneerselvam et al., 2022). Genome annotation was performed using both the NCBI Prokaryotic Genome Annotation Pipeline (PGAP; 2023 release) and the RAST server (Aziz et al., 2008). The prediction of putative gene clusters involved in the production of bioactive secondary metabolites was performed with the antiSMASH server (Blin et al., 2019). Species delimitation and taxonomic assignment were carried out using the Type (Strain) Genome Server (TYGS) (Meier-Kolthoff and Goker, 2019).

2.3. LC-MS/MS analysis

The strain was grown on CCY agar plates at 29 °C for 48 h. An isolated colony was resuspended in 20 μL of 5 × SDS–PAGE loading buffer, heated at 100 °C for 5 min, and analyzed by 11% SDS–PAGE. A major protein band was excised from the gel, placed in 500 μL of ultrapure water, and subjected to liquid chromatography-tandem mass spectrometry (LC–MS/MS) analysis at the proteomics facility of the SCSIE (Servei Central de Suport a la Investigació Experimental), University of Valencia.

Protein identification was performed by peptide mass fingerprinting using the Paragon algorithm (Shilov et al., 2007) implemented in ProteinPilot v5.0 (AB SCIEX, Madrid, Spain). Searches were conducted against the UniProtKB/Swiss-Prot database (Boutet et al., 2007) and BPPRC (Panneerselvam et al., 2022), including curated *B. thuringiensis* reference datasets.

2.4. Insect bioassays

The insecticidal activity of strain Bto_UNVM-42 was evaluated against five insect species: fourth instars of *A. aegypti*, second instars of *A. diaperinus*, neonates of *A. grandis*, neonates of *C. pomonella*, and second-instar larvae of *D. suzukii*. Insects were obtained from laboratory colonies maintained at the Institute of Microbiology and Agricultural Zoology (IMYZA–INTA, Argentina), except for *D. suzukii*, which was reared under controlled laboratory conditions as described below.

For bioassays against *A. aegypti*, spore-crystal suspensions of strain Bto_UNVM-42 were prepared as described by Sauka et al. (2010). Highly concentrated suspensions were adjusted to final concentrations of 100 μg/mL and 1 mg/mL, expressed as dry weight of the spore-crystal biomass, and resuspended in sterile distilled water. The suspensions were then added to plastic cups containing dechlorinated water, following Ibarra et al. (2003) with minor modifications. Twenty fourth-instar larvae were used per treatment, and assays were performed in three independent replicates. Distilled water was used as a negative control. Larval mortality was recorded after 24 h of exposure at 29 °C. Larvae were considered dead when they failed to respond to gentle probing (Sauka et al., 2010).

For bioassays against *A. diaperinus*, *A. grandis* and *C. pomonella*, strain Bto_UNVM-42 was cultured in BM medium. Spore-crystal mixtures were recovered by centrifugation, freeze-dried, and incorporated into the corresponding artificial diets at final concentrations ranging from 5 to 1000 μg/mL, expressed as dry weight of the freeze-dried spore–crystal biomass, following Pérez et al. (2017). Each assay consisted of 24 larvae per treatment and was independently replicated twice. Mortality was recorded after 15 days for *A. diaperinus* and *A. grandis*, and after 5 days for *C. pomonella*. Larvae were considered dead when they failed to respond to gentle probing. Sterile distilled water was used as a negative control (Pérez et al., 2017).

For *D. suzukii*, spores and parasporal crystals were produced by culturing the bacterium in CCY medium supplemented with mineral salts at 29 °C under shaking conditions for 48 h, until complete sporulation. Cultures were harvested and washed sequentially with 1 M NaCl-10 mM EDTA and 10 mM KCl to remove residual medium components. The final pellet was resuspended in sterile distilled water and adjusted to an OD_{600} of 2. Serial dilutions were prepared to obtain suspensions with OD_{600} values of 1 and 0.66. Oral bioassays with *D. suzukii* larvae were performed using the agar-trap method described by Mastore et al. (2021), with minor modifications (Mastore et al., 2021). Briefly, bacterial suspensions were mixed 1:1 (v/v) with cooled agar containing 0.75% (w/v) agar and 5% (w/v) sucrose, yielding final OD_{600} values of 1, 0.5 and 0.33. The mixtures were poured into standard *Drosophila* rearing vials to form a thin agar layer. Thirty second-instar larvae were transferred onto the agar surface in each vial and incubated at 25 °C under a 16:8h light:dark photoperiod. Larval survival and feeding activity were monitored at 24, 48 and 72 h.

2.5. Bioassays with P. redivivus

For assays with the free-living nematode *P. redivivus*, sporulated bacterial lawns were obtained by growing strain Bto_UNVM-42 on BM agar plates (90 × 15 mm) at 29 °C for 72 h, following the qualitative approach described by Sauka et al. (2022). Approximately 4,000 nematodes were placed directly onto the surface of sporulated lawns. Untreated controls consisted of nematodes incubated on BM sporulation agar plates without bacterial material. Nematode viability was assessed after 24 h of exposure at 29 °C, and individuals were considered dead based on the complete absence of movement. In addition, characteristic pathological alterations, including swelling, tissue degradation and loss of pigmentation, were recorded by light microscopy (200 ×) by observing 20 fields per Petri dish, and mortality percentage was determined. Bioassays were performed in triplicate, and assays showing natural mortality in the controls of less than 5% were considered valid. No chemical positive control was included in these assays, as *P. redivivus* was used as a laboratory screening model.

To evaluate the contribution of soluble and cellular fractions, strain Bto_UNVM-42 was cultured in BM sporulation broth at 29 °C for 72 h with agitation until autolysis was observed. Cultures were centrifuged to separate the cellular biomass (pellet) from the cell-free supernatant. The supernatant was collected and used directly without concentration, and the biomass fraction was resuspended in sterile distilled water to the original culture volume. Mortality bioassays were performed by exposing *P. redivivus* individuals to either the biomass fraction or the corresponding cell-free supernatant for 24 h at 29 °C. Five independent microscopic fields per treatment were examined, with 3–6 nematodes per field depending on availability. Individuals were scored as alive or dead after exposure. Mortality was expressed as the proportion of dead nematodes relative to the total number evaluated per treatment. Ninety-five percent confidence intervals (95% CI) were calculated assuming a binomial distribution using the Wilson method.

2.6. Bioassays with plant-parasitic nematodes

Ovicidal and nematicidal activities against plant-parasitic nematodes were evaluated using spore–crystal mixtures at a concentration of 1.075 mg/mL in sterile distilled water, expressed as dry weight of the spore–crystal biomass. The chemical nematicide Vydate® (a.i. oxamyl) at 1000 mg/L was used as a positive control, while sterile water served as the natural mortality control. Nematode inocula were obtained from tomato roots infected with *M. javanica* and from pure cultures of *P. zeae* maintained at the Nematology Laboratory of EEA INTA Balcarce (Argentina).

Egg masses of *M. javanica* were extracted from infected tomato roots and treated with 0.5% (v/v) sodium hypochlorite for 3 min. Eggs were collected by sieving through 2 mm and 50 μm meshes, thoroughly washed with water, and incubated at 25 °C to obtain second-stage juveniles (J2). *P. zeae* inoculum was produced on carrot disks and recovered by washing and decanting through a 50 μm mesh sieve. *In vitro* bioassays were conducted in three-compartment Petri dishes (4 mL per compartment). Each treatment included 25 eggs and 40 J2 larvae of *M. javanica*, or 30 larvae of *P. zeae*. Egg hatching was recorded after 10 days, while larval mortality and immobilization were assessed at 24, 48 and 72 h. Ovicidal activity was expressed as the percentage of hatched larvae, and nematicidal or nematostatic activity as the percentage of dead and/or immobilized nematodes. All quantitative bioassays followed a completely randomized design with three independent replicates per treatment. Mortality and immobilization data were analyzed using one-way analysis of variance (ANOVA), followed by Tukey's *post hoc* test for multiple comparisons ($\alpha = 0.05$). Statistical analyses were performed using R software (version 4.0.0).

2.7. Assessment of β-exotoxin production

The ability of strain Bto_UNVM-42 to produce type I β-exotoxin was assessed by monitoring adult emergence of *Musca domestica*, following the methodology described by Sauka et al. (2014). Strain HD-2, a known β-exotoxin producer, was included as a positive control (Sauka et al., 2014).

3. Results

3.1. Strain isolation and preliminary characterization

The novel *Bacillus* sp. isolate exhibited a typical *B. thuringiensis* phenotype when grown on CCY agar plates, characterized by flat, dry colonies with a matte-white appearance and irregular margins. Light microscopy revealed abundant bipyramidal parasporal crystals stained with Coomassie blue (Fig. 1A). Crystal morphology was further confirmed by scanning electron microscopy (SEM) (Fig. 1B). Based on these characteristic *B. thuringiensis*-like morphological features, the isolate was preliminarily identified as a *B. thuringiensis*-like strain (Vos et al., 2011).

SDS-PAGE analysis of spore–crystal mixtures revealed a predominant protein band with an apparent molecular mass below 130 kDa. This band was excised from the gel and subjected to LC-MS/MS analysis for protein identification (Fig. 1C).

3.2. Genome sequence analysis and annotation

Hybrid Illumina/Oxford Nanopore sequencing yielded a single circular chromosome of 5.43 Mb and seven circular plasmids ranging in size from 11 to 259 kb (Table 1). The chromosome comprises 5,618 predicted coding sequences (CDS) and has a GC content of 35.4%. As observed in other entomopathogenic *Bacillus* species, the genome encodes a complete set of sporulation genes (e.g. *spo0A*, *spoIIE* and *spoV*), multiple proteases and chitinases, as well as several predicted non-ribosomal peptide synthetase/polyketide synthase clusters consistent with siderophore biosynthesis (petrobactin and bacillibactin) (Table 1). Among the plasmids, pBto-42_3 (~181 kb; GC 34.8%; 194 CDS) is distinguished by an extensive array of genes encoding putative pesticidal proteins. Based on manual curation, sequence similarity using custom BLAST searches and BestMatchFinder in the BPPRC database, plasmid pBto-42_3 carries five Cry32-like proteins, two additional Cry-like proteins belonging to the Cry69 and Cry73 families, three Cyt1-like proteins, and two Mpp-like proteins (Table 2, Fig. 2).

Two additional large plasmids encode gene sets that may contribute to host interaction. The ~ 259 kb plasmid (pBto-42_1) contains two thiol-activated cytolysin genes, non-hemolytic enterotoxin A and its lytic component L1, a hemolytic enterotoxin, and regulators of polyketide synthase expression. Notably, the ~ 253 kb plasmid (pBto-42_2) carries two extra chitinases (EC 3.2.1.14), together with several serine and zinc metalloproteases, and a sporulation kinase E (EC 2.7.13.3). The four smaller plasmids primarily encode stability and mobility functions. The ~ 102 kb (pBto-42_4) and the ~ 82 kb (pBto-42_5) plasmids encode partitioning and replication proteins (e.g. ParA, replication initiation proteins) and are enriched in mobile element proteins and transposases.

Table 1
General genomic features of the *B. toyonensis* biovar Thuringiensis strain Bto_UNVM-42. The table summarizes the size, GC content, number of coding sequences (CDS), putative virulence factors, and predicted secondary metabolite clusters identified in the chromosome and each plasmid replicon.

Replicon	Size (bp)	GC (%)	CDS	Virulence factors	Gene Clusters
Chromosome	5,434,080	35.4	5,618	Chitinases, Sep1-like	Petrobactin, bacillibactin
pBto-42_1	259,328	32.6	236	Two thiol-activated cytolysins	–
pBto-42_2	253,526	32.3	279	Two chitinases (EC 3.2.1.14), Sep1-like	–
pBto-42_3	181,212	34.8	194	Cry, Cyt and Mpp family proteins	–
pBto-42_4	102,581	34.7	112	–	–
pBto-42_5	82,696	34.5	117	–	–
pBto-42_6	14,425	33.4	32	–	–
pBto-42_7	11,311	34.4	14	–	–

Table 2
Best-matching reference proteins, amino acid identity and query coverage for the putative pesticidal proteins encoded by plasmid pBto-42_3 of *B. toyonensis* biovar Thuringiensis strain Bto_UNVM-42. For each predicted protein, the best match in the BPPRC database is indicated together with the corresponding percentage of amino acid identity and query coverage. Putative pesticidal proteins are designated using neutral numerical identifiers (e.g., Cry32-1, Cry32-2), reflecting family-level similarity based on sequence comparison and not implying formal pesticidal protein nomenclature.

Locus*	Best match	% identity	% coverage	Assigned family
Cry32-1	Cry32Ea1	46	95	Cry32-like
Cry32-2	Cry32Ga1	44	99	
Cry32-3	Cry32Ka1	53	50	
Cry32-4	Cry32Mb1	41	96	
Cry32-5	Cry32Ua1	41	92	
Cry69-1	Cry69Aa2	43	68	Cry69-like
Cry73-1	Cry73Ba2	33	98	Cry73-like
Cyt1-1	Cyt1Aa1	71	92	Cyt1-like
Cyt1-2	Cyt1Da1	49	97	
Cyt1-3	Cyt1Da2	31	43	
Mpp-1	Mpp3Aa2	29	93	Mpp-like
Mpp-2	Mpp23Aa1	27	77	

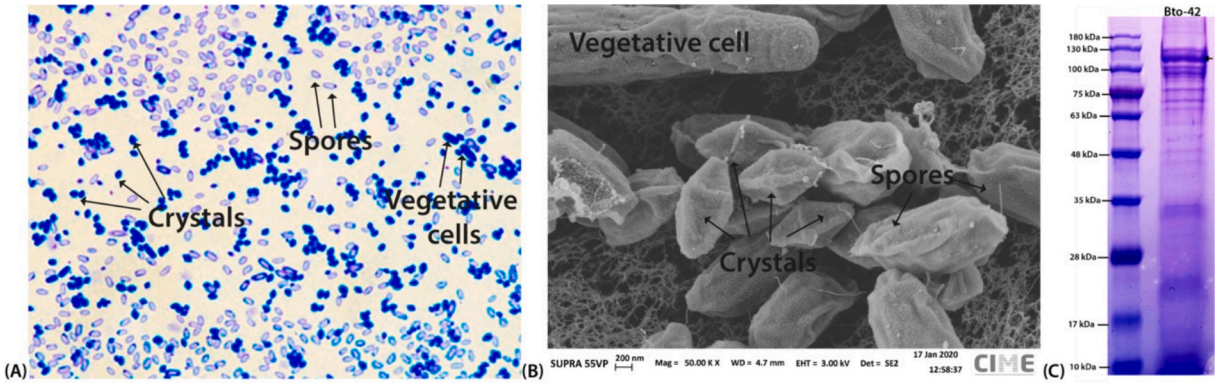

Fig. 1. Microscopic analysis of the *Bacillus* sp. Isolate. (A) Bi-pyramidal parasporal crystals stained with Coomassie blue stain (Ammons et al., 2002), (B) parasporal crystals' examination by SEM analysis and (C) spore-crystal mixtures SDS-PAGE electrophoresis analysis of strain Bto_UNVM-42. Lane 1: molecular weight marker, Lanes 2: spore-crystals mixtures of strain Bto_UNVM-42. The black arrow indicates excised major band containing the Cry32-like proteins. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

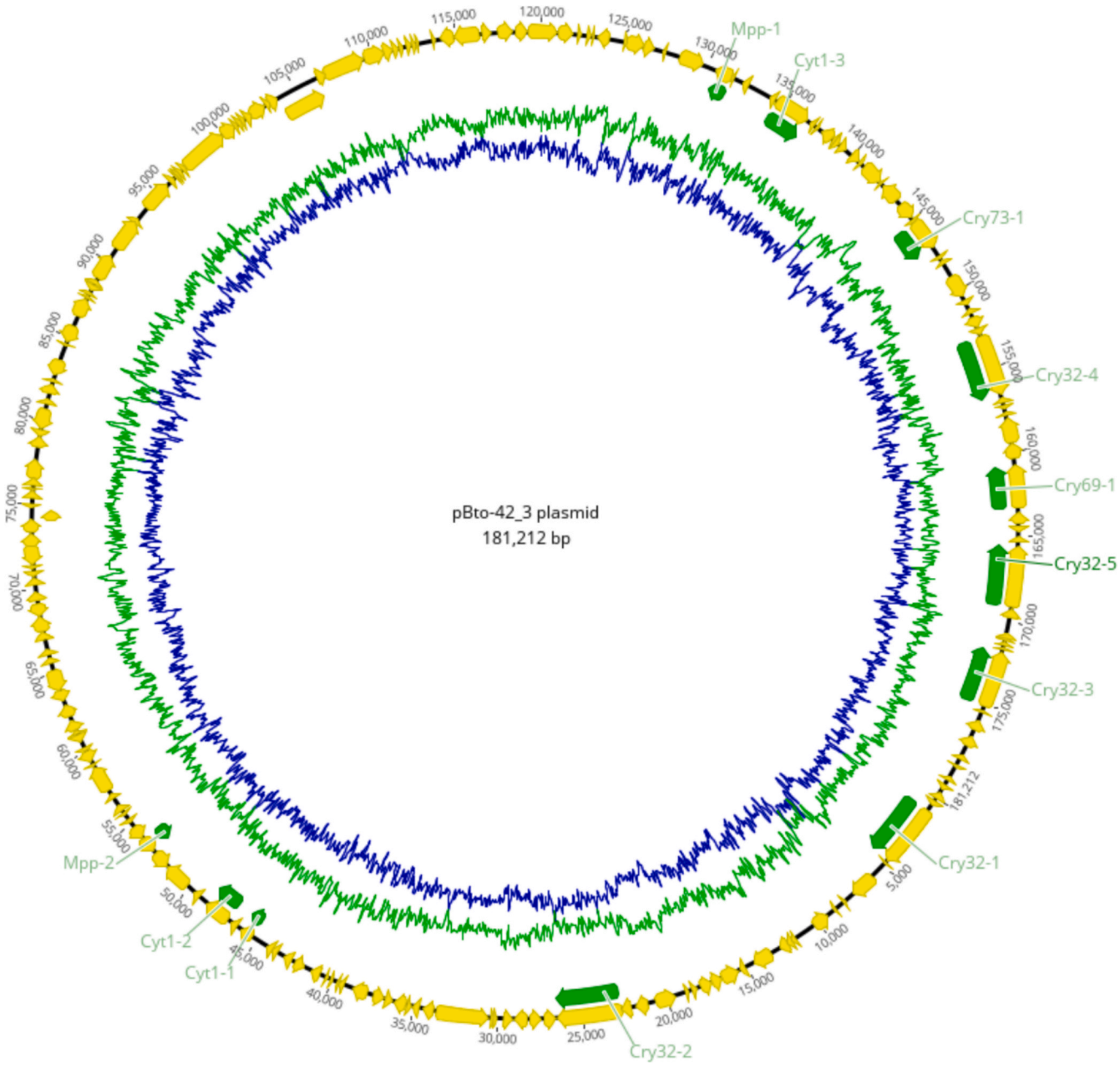

Fig. 2. Circular map of plasmid pBto-42_3 from *B. toyonensis* biovar Thuringiensis strain Bto_UNVM-42. The outer ring shows predicted coding sequences (yellow) and manually annotated pesticidal protein genes (green), including Cry32-like, Cry69-like and Cry73-like proteins, Cyt1-like proteins and Mpp-like proteins. Individual loci are labelled using numerical identifiers (Cry32-1 to Cry32-5, Cry69-1, Cry73-1, Cyt1-1 to Cyt1-3 and Mpp-1 to Mpp-2), which correspond to the predicted proteins listed in Table 2. The inner plots represent GC content (blue) and AT content (green) across the plasmid sequence. The genome size is indicated in the center of the map. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The ~ 14 kb plasmid (pBto-42_6) consists largely of hypothetical proteins with no obvious virulence genes. Finally, the ~ 11 kb plasmid (pBto-42_7) contains a Phd–Doc toxin–antitoxin module, supporting plasmid maintenance.

The Bto_UNVM-42 genome also encodes two CDS located on the chromosome and plasmid pBto-42_2, which exhibit 62% and 51% pairwise amino acid similarity, respectively, to the nematicidal serine protease Sep1 from *B. firmus* DS-1 (Acc. No. EWG10090) (Geng et al., 2016). Based on this finding, we evaluated the hydrolytic activity of Bto_UNVM-42 using casein as a substrate. Our results revealed abundant visible casein hydrolysis, consistent with the presence of predicted proteases in the genome (data not shown). This strong caseinolytic activity was also observed with the Sep1 enzyme from *B. firmus* DS-1 (Geng et al., 2016).

Lastly, genomic screening for the *thu* gene cluster, involved in type I β-exotoxin (thuringiensin) biosynthesis, yielded negative results, with no associated genes identified in the chromosome or plasmids (Sauka et al., 2014).

The phylogenetic tree generated using the Type Strain Genome Server (TYGS) showed that the novel strain clustered uniquely with the type strain *B. toyonensis* NCIMB 14858 (Fig. 3), confirming its classification within the *B. toyonensis* species. In accordance with the genomic-based nomenclature proposed by Carroll et al. (2020) for the *B. cereus* group, the strain is designated as *B. toyonensis* biovar Thuringiensis Bto_UNVM-42. Additionally, the PubMLST server (Jolley et al., 2018) further supported this classification by identifying the genome as

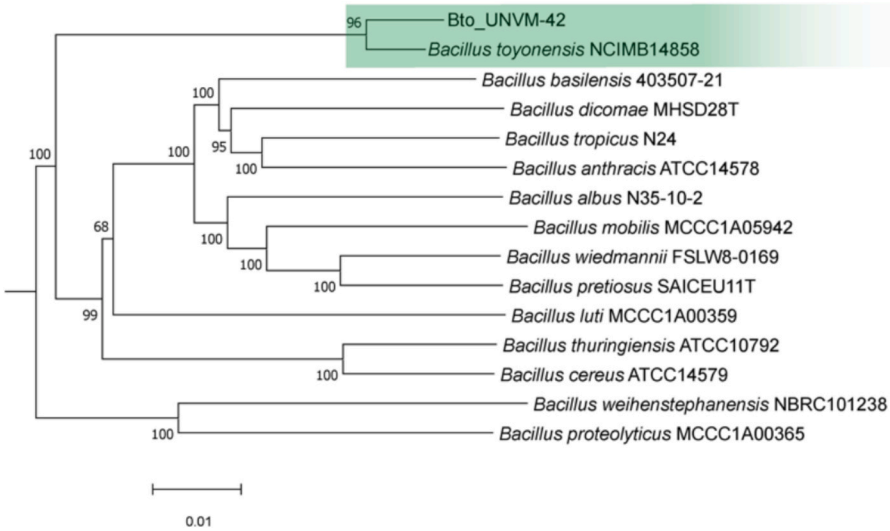

Fig. 3. Whole-genome sequence-based GBDP tree using TYGS server (average branch support 96%) (Meier-Kolthoff and Goker, 2019). The green square highlighted the unique branch containing the *B. toyonensis* type strain NCIMB14858 and the *B. toyonensis* biovar Thuringiensis strain Bto_UNVM-42. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

belonging to the *B. toyonensis* species with 100% support.

3.3. LC-MS/MS analysis

LC-MS/MS analysis of the excised protein band from the SDS-PAGE gel identified multiple peptides corresponding to Cry32-family proteins (Table 3). The detected peptides ranged from 7 to 33 amino acids in length and showed moderate to high spectral counts, indicating robust proteomic evidence. Based on the LC-MS/MS search engine assignments, all peptides were associated with Cry32-family entries in the searched database. The occurrence of peptide matches to different Cry32-annotated entries reflects the high sequence similarity among Cry32-family members and the resulting ambiguity of peptide-level identification.

3.4. Bioassays

Mixed spore-crystal suspensions of strain Bto_UNVM-42 exhibited no toxic activity against *A. aegypti*, *A. diaperinus*, *A. grandis*, *C. pomonella* and *D. suzukii*. In contrast, the strain caused low but detectable mortality (~8%) in *M. javanica* J2 larvae. The chemical control Vydate® induced ~ 23% mortality and showed significantly higher mortality than both Bto_UNVM-42 and the water control ($p < 0.05$), whereas no significant differences were detected between Bto_UNVM-42 and water. Bto_UNVM-42 did not significantly affect the percentage of hatched larvae compared to the water control. In contrast, Vydate® showed a significantly different response compared to both treatments ($p < 0.05$), consistent with the statistical analysis reported in the original assay. For *P. zeae*, the progression of larval immobilization from 24 to 72 h revealed significant differences among treatments. Vydate® induced approximately 65% immobilization, whereas the highest effect was observed with the Bto_UNVM-42 strain, reaching 86% immobilization ($p < 0.05$). After completion of the bioassay, treated larvae were washed and transferred to sterile water. While larvae exposed to Vydate® did not recover mobility, those treated with Bto_UNVM-42 progressively regained movement. This indicates that, in *P. zeae*, the Bto_UNVM-42 strain acts as a nematostatic agent, inhibiting larval movement during exposure but allowing recovery once the treatment is removed. Replicate-level data for the *M. javanica* and *P. zeae* bioassays are provided in Supplementary Table S1.

Exposure of *P. redivivus* to sporulated bacterial lawns of strain Bto_UNVM-42 grown on BM agar plates resulted in complete mortality within 24 h at 29 °C. Treated nematodes exhibited characteristic pathological alterations, including swelling, tissue degradation, and loss of pigmentation (Fig. 4). In contrast, when 72 h liquid cultures were fractionated, no mortality was observed in either the untreated control (0/25) or the biomass fraction (0/20). The corresponding cell-free supernatant induced mortality in 6 out of 27 evaluated nematodes, corresponding to 22.2% mortality (95% CI: 10.6–40.8%). Mortality in the supernatant treatment was unevenly distributed among observation fields (Supplementary Table S1), reflecting variability at the observational level within a single culture-derived treatment.

Table 3
LC–MS/MS identification of Cry-family peptides in strain Bto_UNVM-42. Peptides identified from the excised SDS-PAGE protein band and assigned to Cry32-family proteins based on LC–MS/MS database annotation.

Peptide (LC-MS/MS)	Peptide length	Spectral count	LC–MS/MS assigned protein
ALFTNDAK	8	1	Cry32Ea1
AVNALFTDDAK	11	4	Cry32Ea1
AVNALFTNDAK	11	2	Cry32Ea1
DMTLMVLDIVAVWPTYNPRIYTNPNGVQVELSR	33	1	Cry32Ma1
GPGSTGGDLVK	11	21	Cry32Ga1
IASENGVANIDNVEIIEAHPLTGEALAR	28	2	Cry32Ea1
KAVNALFTNDAK	12	18	Cry32Ea1
MNLNDNK	7	3	Cry32Ma1
TLEIFPETDR	10	18	Cry32Ma1

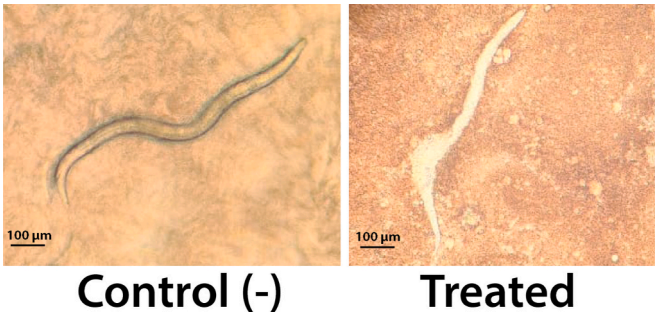

Fig. 4. The picture shows control and treated *P. redivivus* larvae with strain Bto_UNVM-42 (20×). Treated larvae appeared swollen, degraded and opaque.

3.5. β-exotoxin bioassay

Strain Bto-UNVM-42 was classified as negative for β-exotoxin (thuringiensin) production based on the *M. domestica* bioassay, as adult emergence was observed. In addition, and consistent with the lack of teratological effects of strain Bto-UNVM-42 on *M. domestica*, no genes belonging to the *thu* gene cluster involved in type I β-exotoxin biosynthesis were identified in the genome (Sauka et al., 2014).

4. Discussion

B. thuringiensis is a gram-positive, soil-dwelling bacterium widely recognized for its role as a biocontrol agent against a broad range of insect pests. This bacterium produces crystalline proteins during sporulation, known as δ-endotoxins or crystal proteins, which are toxic to specific invertebrates upon ingestion (Jouzani et al., 2017). The nematicidal activity of several Cry proteins, namely Cry5, Cry12, Cry13, Cry14, Cry21, Cry55 and App6 (formerly Cry6), has been well documented in the scientific literature (Berry et al., 2025; van Frankenhuyzen, 2009; van Frankenhuyzen and Tonon, 2013). The extensive use of *B. thuringiensis*-based products in agriculture has contributed significantly to pest suppression, crop protection and the promotion of more sustainable agricultural practices. In addition, the genetic engineering of crops expressing *B. thuringiensis* pesticidal proteins has provided an effective tool within integrated pest management strategies (Kumar et al., 2008).

Beyond *B. thuringiensis*, other members of the *Bacillus cereus* group have been reported to produce parasporal crystals with biocidal activity, including *Bacillus wiedmannii* (Lazarte et al., 2018) and *B. toyonensis* biovar Thuringiensis (Sauka et al., 2022). The latter biovar was distinguished by the production of insecticidal proteins with dual activity against *C. pomonella* and *A. grandis*, while showing reduced activity against *A. aegypti* and *A. diaperinus*. Importantly, this strain did not display toxicity against the free-living nematode *P. redivivus*, suggesting a primarily insect-targeted specificity.

The classification of Bto_UNVM-42 as *B. toyonensis* biovar Thuringiensis is fundamentally based on its ability to produce parasporal crystals, a phenotypic hallmark of the Thuringiensis biovar within the *B. cereus* group. Proteomic analysis identified Cry32-family proteins as the major components of these crystals. Members of this family have been reported to be active against *Conopomorpha cramerella* (Lepidoptera) and *Myzus persicae* (Hemiptera) (Berry et al., 2025). In the present study, the parasporal crystals showed no activity against the Lepidoptera, Coleoptera, and Diptera species tested, which is consistent with the known target spectrum of Cry32 proteins. Therefore, the insecticidal potential of this strain cannot be fully discarded without testing against additional hemipteran or other sucking insect pests.

The complete mortality observed when *P. redivivus* was exposed to sporulated lawns suggests that nematicidal activity is markedly enhanced under solid culture conditions. In contrast, fractionation of 72 h liquid cultures revealed no detectable activity in the biomass fraction and modest mortality in the corresponding cell-free supernatant. This difference may reflect the combined effects of sporulation-associated factors and culture-dependent accumulation of active components on solid substrate. Under liquid conditions, dilution effects, altered physiological states, or reduced stability of active compounds may limit the observed activity. Although the precise contribution of crystal proteins, spores, and soluble factors cannot be determined based on the present data, these findings indicate that nematicidal expression by strain Bto_UNVM-42 may also be influenced by growth conditions.

Notably, the morphology of *P. redivivus* cadavers exposed to Bto_UNVM-42, characterized by swelling, loss of pigmentation and extensive tissue degradation (Fig. 4), is consistent with a proteolytic or enzymatic mechanism. Similar pathological features have been reported for *Bacillus nematocida*, which produces the extracellular serine protease Bace16 that degrades the nematode cuticle of *P. redivivus* (Niu et al., 2007). Comparable effects have also been described for the gram-negative bacterium *Photorhabdus laumondii* subsp. *laumondii*, which induces visible cuticle ruptures and degradation near the basal bulb of nematodes exposed to aged bacterial cultures (Salas et al., 2021). The observed pathological effects on *P. redivivus*, such as cuticle degradation and swelling, differ from the classical pore-forming intoxication of nematicidal Cry proteins. Instead, these symptoms align with an enzymatic mode of action. Although Cry32 proteins were the major components identified in the crystals, their specific role in nematode mortality remains unclear. Given the presence of Sep1 homologs and the absence of known nematicidal *cry* genes, our data are consistent with the hypothesis that Bto_UNVM-42 may exert its effects predominantly through an exogenous enzymatic attack on the nematode integument rather than through typical crystal-mediated midgut disruption. A more comprehensive proteomic analysis of the complete spore-crystal mixture, as well as of culture supernatants, would be required to clarify the molecular basis of this activity and help to determine whether Sep1 homolog proteins or other secreted or surface-associated proteins are directly involved in nematode killing.

The differential activity observed among nematode species may also be influenced by differences in feeding behavior and cuticle structure. *P. redivivus* is a free-living bacterivorous nematode, which directly feeds on bacterial cells and may therefore be more readily exposed to surface-associated or secreted bacterial factors. In contrast, plant-parasitic nematodes such as *M. javanica* and *P. zeae* exhibit distinct feeding strategies and possess structural adaptations of the cuticle and stylet apparatus that may limit direct exposure to bacterial components. These biological differences may contribute to the variation in susceptibility observed in our assays.

In addition, strain Bto_UNVM-42 did not inhibit egg hatching but induced low yet detectable mortality (~8%) in second-stage juveniles (J2) of *M. javanica*. More strikingly, it produced a significant but reversible nematostatic effect on *P. zeae* juveniles. A similar reversible nematostatic effect has been reported for *M. incognita* treated with BioNem WP 5, a commercial formulation based on spore–crystal mixtures of *B. firmus* strain I-1582 (Terefe et al., 2009). This strain has been approved for use in Europe and constitutes the active ingredient of other environmentally friendly nematicidal products such as Flocter (Ghahremani et al., 2020).

In this context, it is particularly relevant that strain Bto_UNVM-42 encodes two homologs of the cuticle-degrading serine protease Sep1 from *B. firmus*, sharing 51% and 62% pairwise amino acid similarity. *B. firmus* DS-1 has been shown to exert nematicidal activity against *Caenorhabditis elegans* and *M. incognita* through Sep1-mediated degradation of the nematode cuticle (Geng et al., 2016). Although the presence of Sep1 homologs does not demonstrate functional equivalence, their detection supports the possibility of a related enzymatic mechanism.

Although this proposed mode of action requires further experimental validation, particularly in additional plant-parasitic nematodes, the reversible nematostatic effect observed against *P. zeae* should be

interpreted cautiously but remains biologically significant and warrants further investigation.

5. Conclusions

This study reports, for the first time, a *B. toyonensis* biovar Thuringiensis strain exhibiting nematicidal activity, thereby expanding the known functional diversity of this species and its potential applications in biological control. Unlike classical *B. thuringiensis* nematicidal systems, which rely on the ingestion of specific Cry proteins, the activity observed for strain Bto_UNVM-42 may involve an alternative mechanism. The absence of known *B. thuringiensis*-like nematicidal proteins, together with the presence of Sep1-like serine protease homologs and the characteristic pathological effects observed in exposed nematodes, is consistent with the possibility of a predominantly enzymatic mode of action, potentially involving cuticle degradation rather than toxin-mediated intoxication.

The reversible nematostatic effect observed against *P. zeae* further suggests a non-lethal but biologically relevant interaction that effectively compromises nematode mobility. Because motility is essential for host invasion, feeding, and reproduction in plant-parasitic nematodes, such an effect could substantially impair their capacity to infest roots and complete their life cycle. While this mode of action requires further experimental validation, particularly in additional phytopathogenic nematode species, it represents a promising and largely unexplored strategy for nematode management.

Future research should focus on elucidating the molecular basis of this activity, clarifying the specific contribution of the Sep1 homologs, and determining the agronomic relevance of the observed effects in terms of dosage, persistence, delivery strategies and target specificity. Overall, these findings highlight *B. toyonensis* biovar Thuringiensis Bto_UNVM-42 as a potential candidate source of novel enzymatic tools for integrated pest management, offering a sustainable alternative or complement to conventional chemical nematicides with reduced environmental impact.

Additionally, future studies should include bioassays against other lepidopteran and hemipteran species to clarify the biological role of the Cry32-family proteins present in the parasporal crystals produced by this strain and to determine their potential within a broader pest management context.

Data availability statement

This whole-genome shotgun project has been deposited in DDBJ/ENA/GenBank under the accession number JAYEPS000000000. The version described in this paper is the first version, JAYEPS000000000.1. The original replicate-level datasets for plant-parasitic nematodes (*M. javanica* and *P. zeae*) and for the free-living nematode *P. redivivus*, together with the official statistical report issued by INTA–EEA Balcarce, have been deposited in Zenodo (https://doi.org/10.5281/zenodo.18722129).

CRediT authorship contribution statement

Diego Herman Sauka: Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Cecilia Peralta:** Methodology, Investigation. **Baltasar Escriche:** Investigation. **Tadeo Fernandez-Göbel:** Methodology, Investigation. **Federico Ocampo:** Methodology, Investigation. **Marino Santos:** Methodology, Investigation. **Augusto Salas:** Investigation. **Eleodoro E. Del Valle:** Methodology, Investigation. **Leopoldo Palma:** Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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7. Key Contribution

This study presents the first report on the isolation and characterization of a previously undescribed *Bacillus toyonensis* biovar Thuringiensis strain, Bto_UNVM-42, carrying putative pesticidal genes (*cry*, *cyt* and *mpp*) and expressing Cry32-family proteins. The strain exhibits strong nematicidal activity against *P. redivivus*, limited activity against *M. javanica*, and nematostatic effects against *P. zeae*.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jip.2026.108595.

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